

CS 330 - Operating Systems

RAID

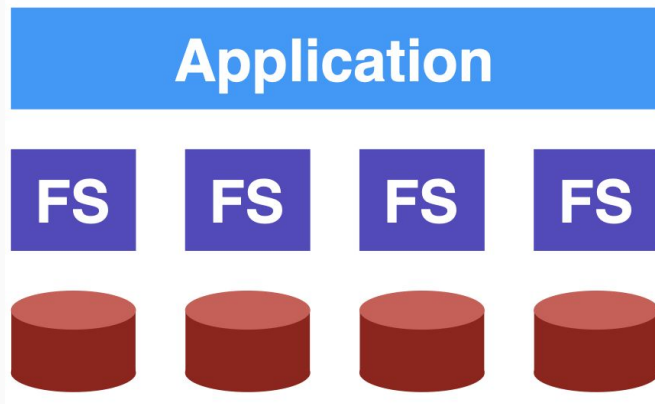
17-11-2025

We want more!

- Want more than one disks
 - Capacity - disks also have limited storage
 - Performance - multiple disk can have parallel I/O
 - Reliability - what if a disk become faulty
- Challenge
 - File systems are mostly built for one disk

JBOD

- Application is smart, stores different files on different file systems
- Starts storing files in other disk once the first disk is full
- Just a bunch of disks!

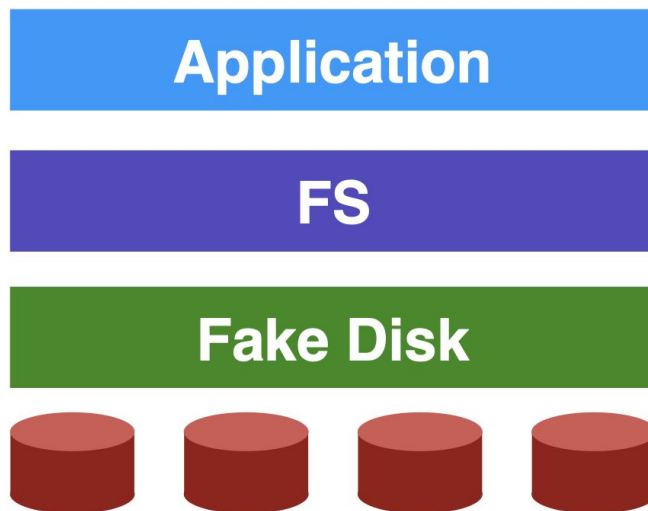


RAID

- Redundant Array of Inexpensive Disks
- Build logical disk from many physical disks

RAID is:

- transparent
- deployable

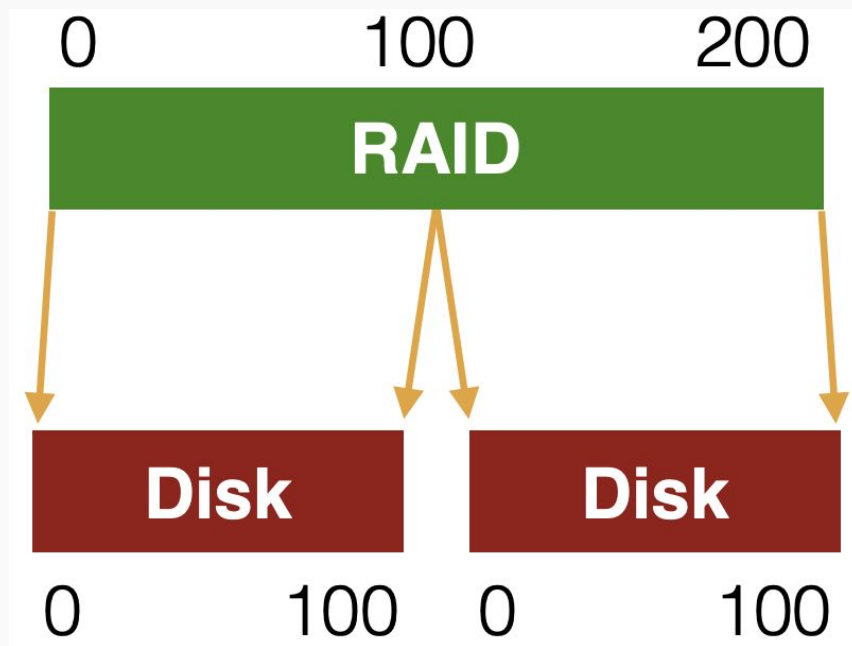


Logical disk gives

- capacity
- performance
- reliability

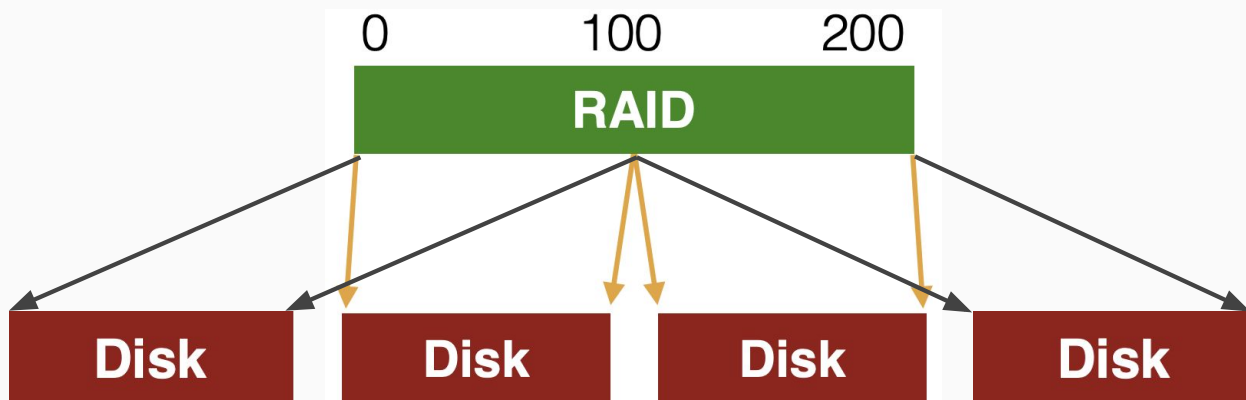
RAID

- Build a fast and large disk from smaller disks



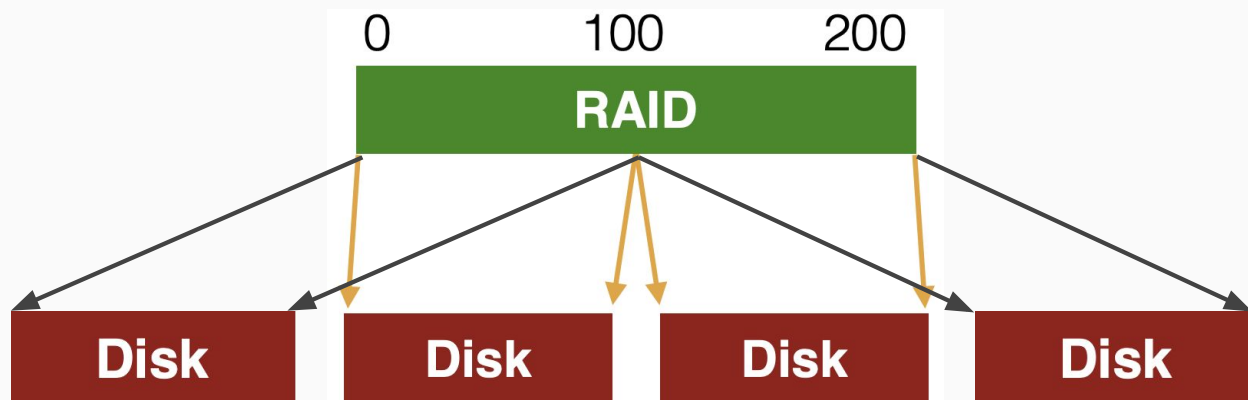
Reliability in RAID

- Add redundancy in data stored on disks



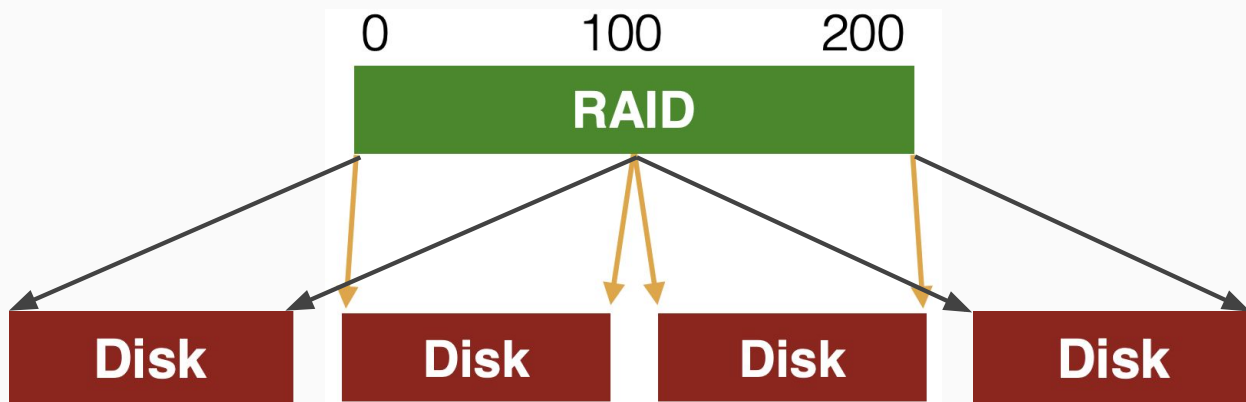
Reliability in RAID

- Add redundancy in data stored on disks
- How many physical copies should RAID keep for every logical block?



Mapping in RAID

- How should we map logical to physical addresses?



Mapping Data

- Static mapping
 - Paging in virtual memory uses *dynamic mapping*
- RAID is fixed in size and available as a separate H/W
- Worry about copying data into multiple disks

Reliability in RAID

- Fail-stop fault model
 - Disks are either working or failed
 - Cannot access any data from failed disks
 - Failed disks are immediately known to the RAID controller

RAID Evaluation

- Capacity
 - With N disks having B blocks, what is the **useful** capacity available to the users?

RAID Evaluation

- Capacity
 - With N disks having B blocks, what is the **useful** capacity available to the users?
 - Without any redundancy, all $N * B$ blocks are available to the user
 - With one copy of each, $N * B / 2$ blocks
 - With some RAIDs, it can be in between too

RAID Evaluation

- Reliability
 - How many disk faults can be tolerated?

RAID Evaluation

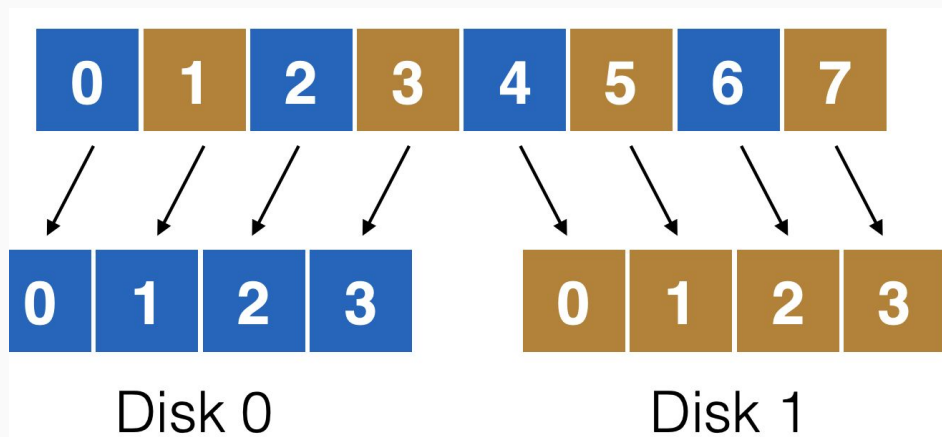
- Reliability
 - How many disk faults can be tolerated?
 - With n mirrors, n disk faults can be tolerated
 - May be different if parts of disks fail

RAID Evaluation

- Performance
 - Depending on the workload, different RAIDs have different performance!

RAID 0 : Striping

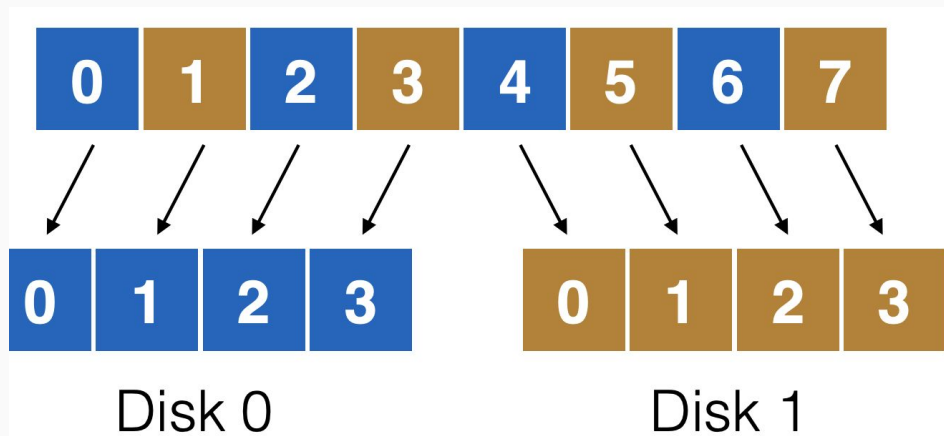
- No redundancy (not exactly a RAID)
- Gives good performance and capacity
- Stripes blocks across the disks
 - Why?



<u>Disk 0</u>	<u>Disk 1</u>
0	1
2	3
4	5
6	7

RAID 0 : Striping

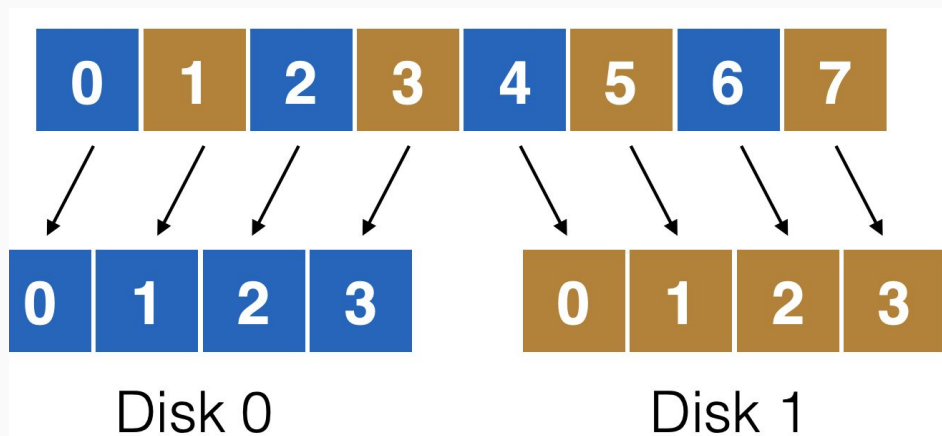
- No redundancy (not exactly a RAID)
- Gives good performance and capacity
- Stripes blocks across the disks
 - Helps parallelize access to blocks



<u>Disk 0</u>	<u>Disk 1</u>
0	1
2	3
4	5
6	7

RAID 0 : Striping

- Access the logical block 4
 - Disk: $4 \% 2$ (logical block $\%$ no. of disks)
 - Offset: $4 / 2$ (logical block $/$ no. of disks)



<u>Disk 0</u>	<u>Disk 1</u>
0	1
2	3
4	5
6	7

RAID 0

- Chunk size is important too
 - Depends on the application
- Stored in chunks of two

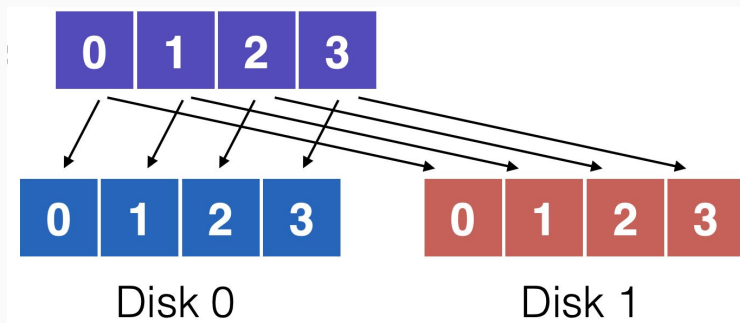
Disk 0	Disk 1	Disk 2	Disk 4
0	2	4	6
1	3	5	7
8	10	12	14
9	11	13	15

RAID 0

- Capacity?
 - $N * B$
- How many disks can fail?
 - 0
- Performance / throughput
 - Sequential read/write - $N * S$ (S is disk transfer rate)
 - Random - $N * R$ (R is disk transfer rate)
- Latency?
 - Read : T / Write : T

RAID 1

- Mirroring
 - Multiple copies of data
 - Disks are fail-stop (work or not)



RAID 1

- Mirroring and Striping (RAID 10)
 - Multiple copies of data
 - Disk 0 and 1 store the same data
 - Disk 2 and 3 store the same data
 - But no common data between 0 and 2

Disk 0	Disk 1	Disk 2	Disk 3
0	0	1	1
2	2	3	3
4	4	5	5
6	6	7	7

RAID 1

- Capacity?
 - $N * B / 2$
- Fail-disks?
 - $N/2$
- Performance?
 - Sequential read and write - $N / 2 * S$ MB/sec
 - Random write - $N / 2 * R$ MB/sec
 - Random read - $N * R$ MB/sec
- Latency?
 - Read : T / Write : T

Disk 0	Disk 1	Disk 2	Disk 3
0	0	1	1
2	2	3	3
4	4	5	5
6	6	7	7

Managing Crashes

- Write(A) to 2
- Write(T) to 3

	Disk0	Disk1
0	A	A
1	B	B
2	C	C
3	D	D

	Disk0	Disk1
0	A	A
1	B	B
2	A	A
3	D	D

	Disk0	Disk1
0	A	A
1	B	B
2	A	A
3	D	T

Managing Crashes

- Crash happens
- Which disk has the correct data?

	Disk0	Disk1
0	A	A
1	B	B
2	A	A
3	D	T

Managing Crashes

- Crash happens
- Which disk has the correct data?
- Consistent-Update Problem
 - Use non-volatile RAM in RAID controller

	Disk0	Disk1
0	A	A
1	B	B
2	A	A
3	D	T

Disk Transfer Rate

Assume a sequential transfer of size 10 MB on average, and a random transfer of 10 KB on average.

Average seek time	7 ms
Average rotational delay	3 ms
Transfer rate of disk	50 MB/s

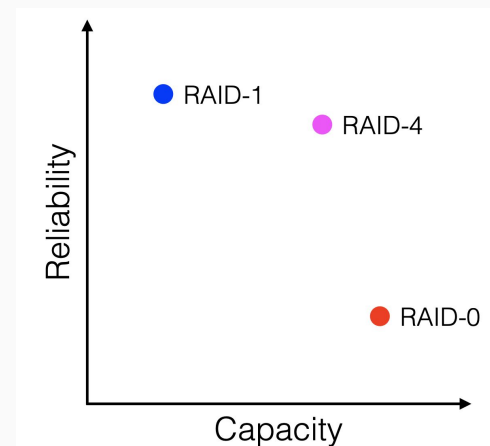
What is:

S (sequential transfer rate) and
R (random transfer rate)?

RAID 4

- Parity check
 - Special parity disk
- Use less capacity with less performance

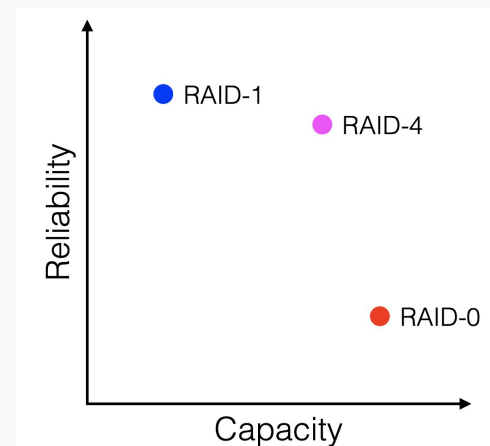
Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
4	5	6	7	P1
8	9	10	11	P2
12	13	14	15	P3



RAID 4

- Mathematical function to recover block loss

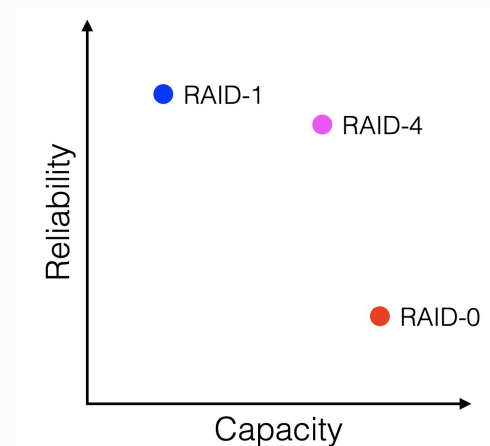
Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
4	5	6	7	P1
8	9	10	11	P2
12	13	14	15	P3



RAID 4

- Mathematical function to recover block loss
 - XOR all bits in the stripe and store it in the parity disk block
 - If N-1 blocks are intact, it is easy to recover the missing block

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
4	5	6	7	P1
8	9	10	11	P2
12	13	14	15	P3



RAID 4

Disk0

Disk1

Disk2

Disk3

Disk4

0

1

0

1

$\text{XOR}(0,1,0,1)=0$

RAID 4

Disk0

Disk1

Disk2

Disk3

Disk4

00

01

10

11

$(\text{XOR}(0,0,1,1),$
 $\text{XOR}(0,1,0,1))=00$

RAID 4

- Capacity?
 - $(N-1) * B$
- How many disks can fail?
 - 1
- Throughput?
 - Sequential write — $(N-1)*S$ (**full stripe write**)
 - Sequential read — $(N-1)*S$
 - Random read — $(N-1)*R$
 - Random write?

RAID 4

- Random writes
 - Additive parity
 - Read all other blocks and XOR with new block
 - Subtractive parity

Disk0

Disk1

Disk2

Disk3

Disk4

0

1

0

1

$\text{XOR}(0, 1, 0, 1) = 0$

(parity)

RAID 4

- Write 0 to Disk 1
 - Read old value of Disk 1
 - Read old value of parity
 - If New value of Disk 1 == Old value of Disk 1, Do nothing
 - Else, Write new flipped parity and Write new value to Disk 1
 - Each random write, needs 2 reads and 2 writes

Disk0	Disk1	Disk2	Disk3	Disk4
0	1	0	1	$\text{XOR}(0, 1, 0, 1) = 0$
				(parity)

RAID 4

- What about multiple writes?
 - All wait to read and write Parity Disk
 - Small-write problem
 - Throughput is $R/2$

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
*4	5	6	7	+P1
8	9	10	11	P2
12	*13	14	15	+P3

RAID 4

- Capacity?
 - $(N-1) * B$
- How many disks can fail?
 - 1
- Throughput?
 - Sequential write — $(N-1)*S$ (**full stripe write**)
 - Sequential read — $(N-1)*S$
 - Random read — $(N-1)*R$
 - Random write — $1/2*R$
- Latency?
 - Read : T / Write : $2T$

RAID 5

- Rotating parity
 - Improve random write performance

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
5	6	7	P1	4
10	11	P2	8	9
15	P3	12	13	14
P4	16	17	18	19

RAID 5

- Capacity ?
 - $(N-1) * B$
- Fail-disks?
 - 1
- Throughput?
 - sequential reads? $(N-1) * S$
 - sequential writes? $(N-1) * S$
 - random reads? $N * R$
 - random writes? $N * R / 4$
- Latency?
 - Read : T / Write : $2T$

Comparison in RAIDs

	RAID-0	RAID-1	RAID-4	RAID-5
Capacity	$N \cdot B$	$(N \cdot B)/2$	$(N - 1) \cdot B$	$(N - 1) \cdot B$
Reliability	0	1 (for sure) $\frac{N}{2}$ (if lucky)	1	1
Throughput				
Sequential Read	$N \cdot S$	$(N/2) \cdot S^1$	$(N - 1) \cdot S$	$(N - 1) \cdot S$
Sequential Write	$N \cdot S$	$(N/2) \cdot S^1$	$(N - 1) \cdot S$	$(N - 1) \cdot S$
Random Read	$N \cdot R$	$N \cdot R$	$(N - 1) \cdot R$	$N \cdot R$
Random Write	$N \cdot R$	$(N/2) \cdot R$	$\frac{1}{2} \cdot R$	$\frac{N}{4} R$
Latency				
Read	T	T	T	T
Write	T	T	$2T$	$2T$

Thank you all!